C4-4a: Create Multiple Tables

Advanced SQL

Entity Relation (ER) Diagram



Linked Entity

Table Description :

Customers (Id, FirstName, LastName, City, Country, Phone)

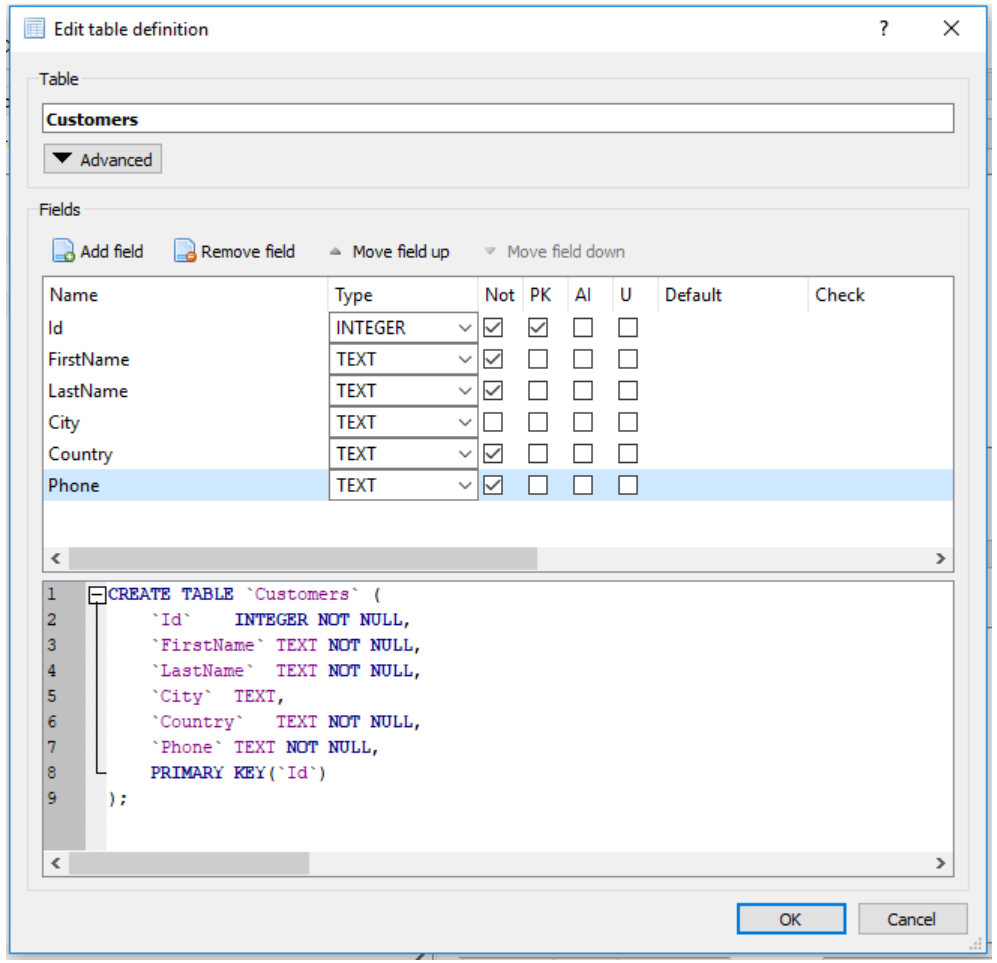
Products(Id, ProductName, UnitPrice, Package, SupplierName, SupplierCountry)

Orders (Id, OrderDate, CustomerID*, ProductID*, Quantity)

__ : Primary Key;

* : Foreign Key

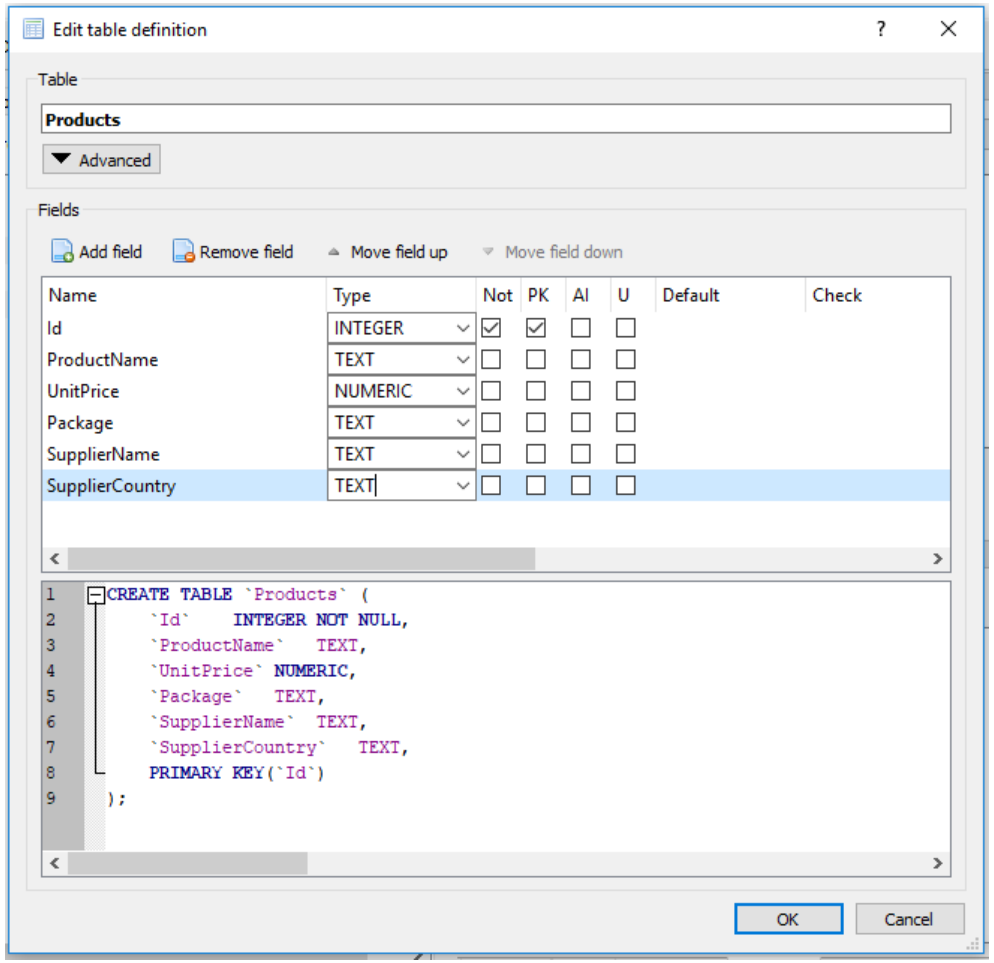
Create Table Customers



Customers
Id
FirstName
LastName
City
Country
Phone

```
CREATE TABLE Customers (
        Id              INTEGER NOT NULL,
        FirstName       TEXT NOT NULL,
        LastName        TEXT NOT NULL,
        City            TEXT,
        Country         TEXT NOT NULL,
        Phone           TEXT NOT NULL,
        PRIMARY KEY(Id)
);
```

Create Table Products

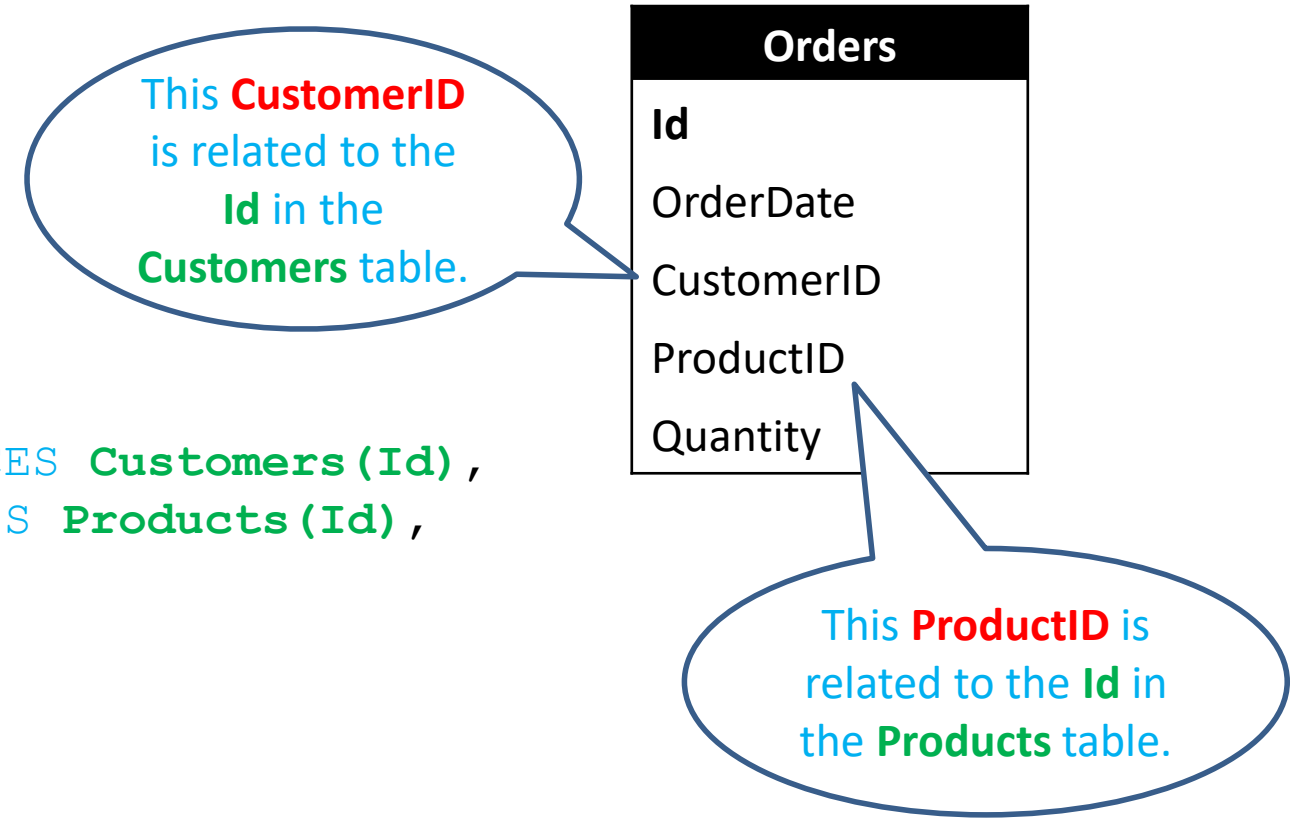


Products
Id
ProductName
UnitPrice
Package
SupplierName
SupplierCountry

```
CREATE TABLE Products (
        Id                      INTEGER NOT NULL,
        ProductName             TEXT,
        UnitPrice               NUMERIC,
        Package                 TEXT,
        SupplierName            TEXT,
        SupplierCountry         TEXT,
        PRIMARY KEY(Id)
);
```

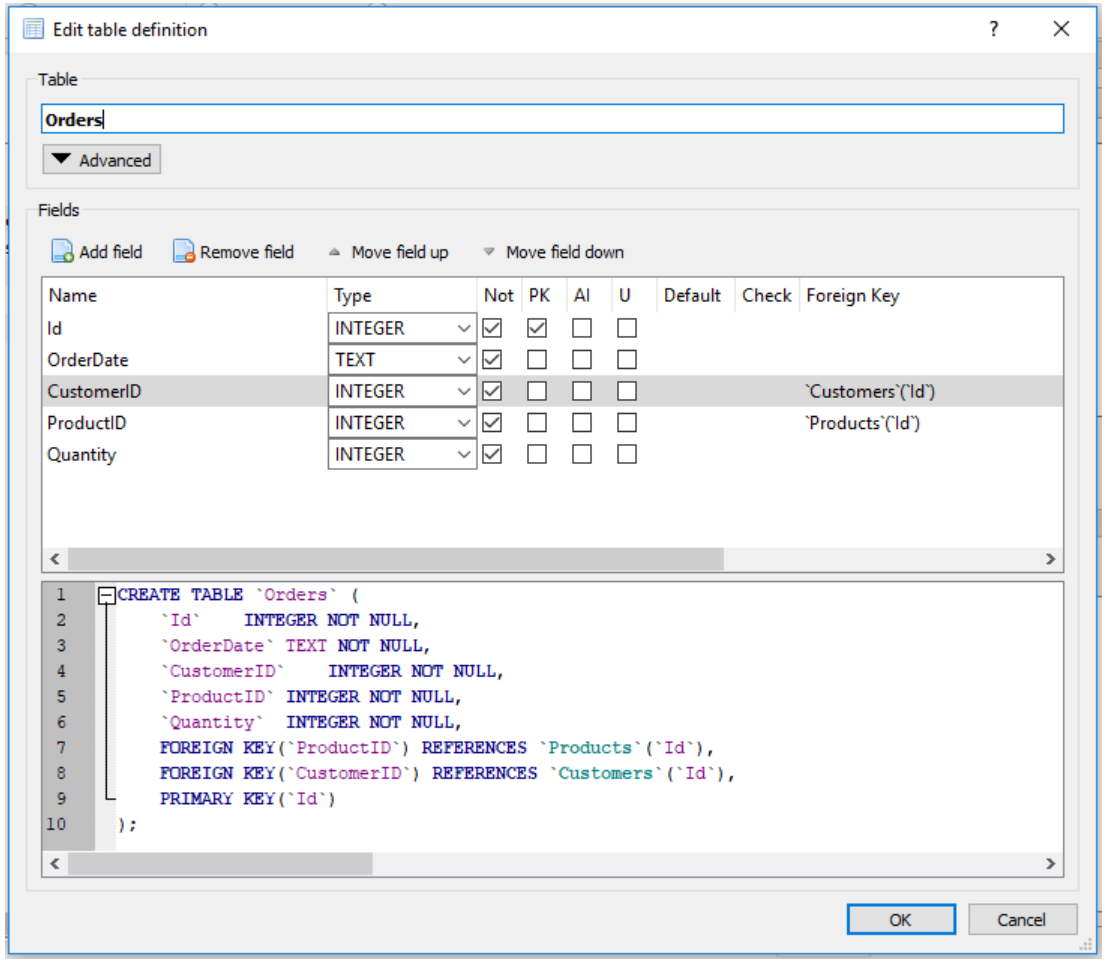
Orders *is a linked entity between* Customers and Products

```
CREATE TABLE Orders (
        Id              INTEGER NOT NULL,
        OrderDate       TEXT NOT NULL,
        CustomerID      INTEGER NOT NULL,
        ProductID       INTEGER NOT NULL,
        Quantity        INTEGER NOT NULL,
        FOREIGN KEY(CustomerID) REFERENCES Customers(Id),
        FOREIGN KEY(ProductID) REFERENCES Products(Id),
        PRIMARY KEY(Id)
);
```



Create Table Orders

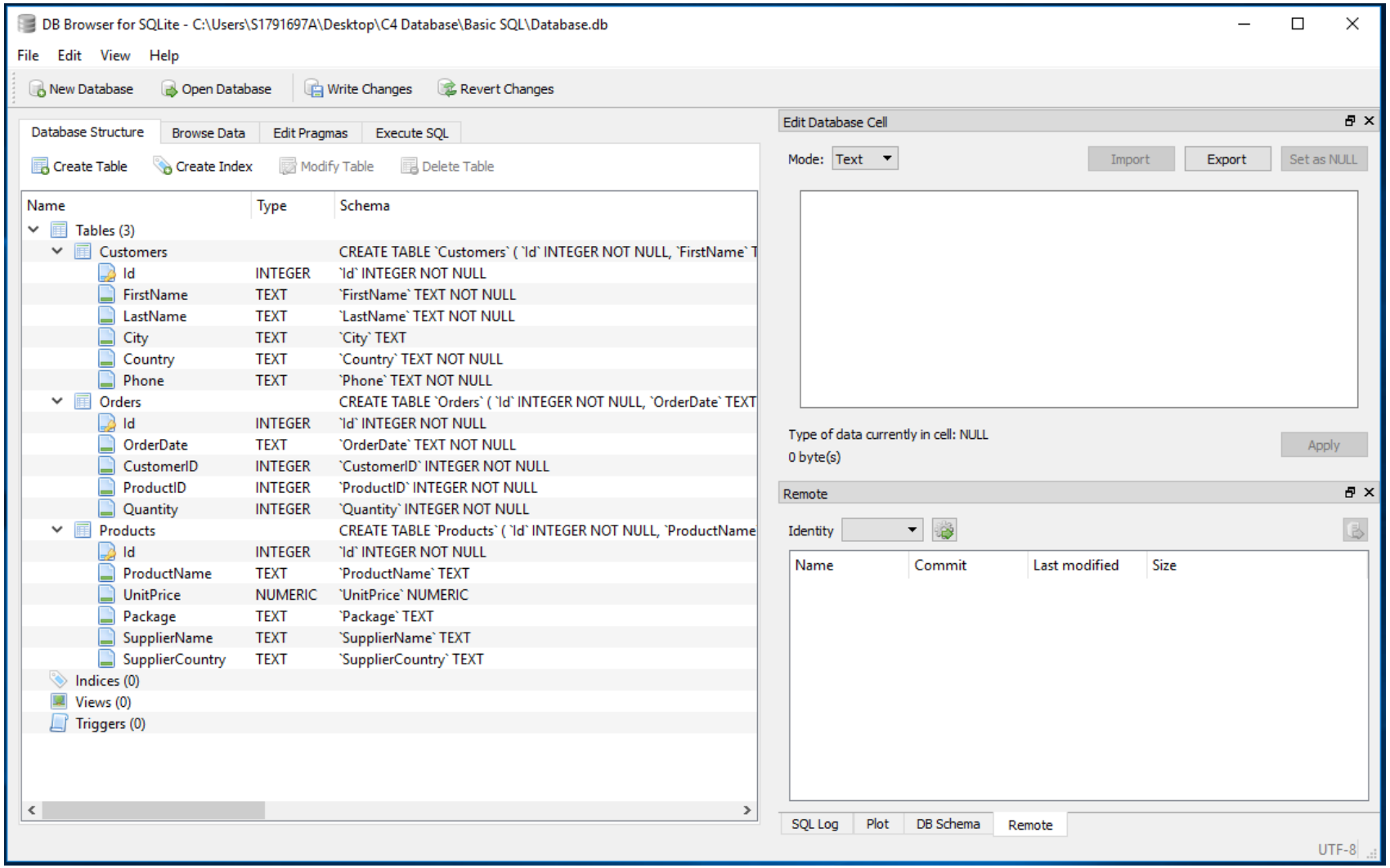
Orders
Id
OrderDate
CustomerID*
ProductID*
Quantity




```
CREATE TABLE Orders (
        Id              INTEGER NOT NULL,
        OrderDate       TEXT NOT NULL,
        CustomerID      INTEGER NOT NULL,
        ProductID       INTEGER NOT NULL,
        Quantity        INTEGER NOT NULL,
        FOREIGN KEY(ProductID) REFERENCES Products(Id),
        FOREIGN KEY(CustomerID) REFERENCES Customers(Id),
        PRIMARY KEY(Id)
);
```

Database Structure – Table Schema

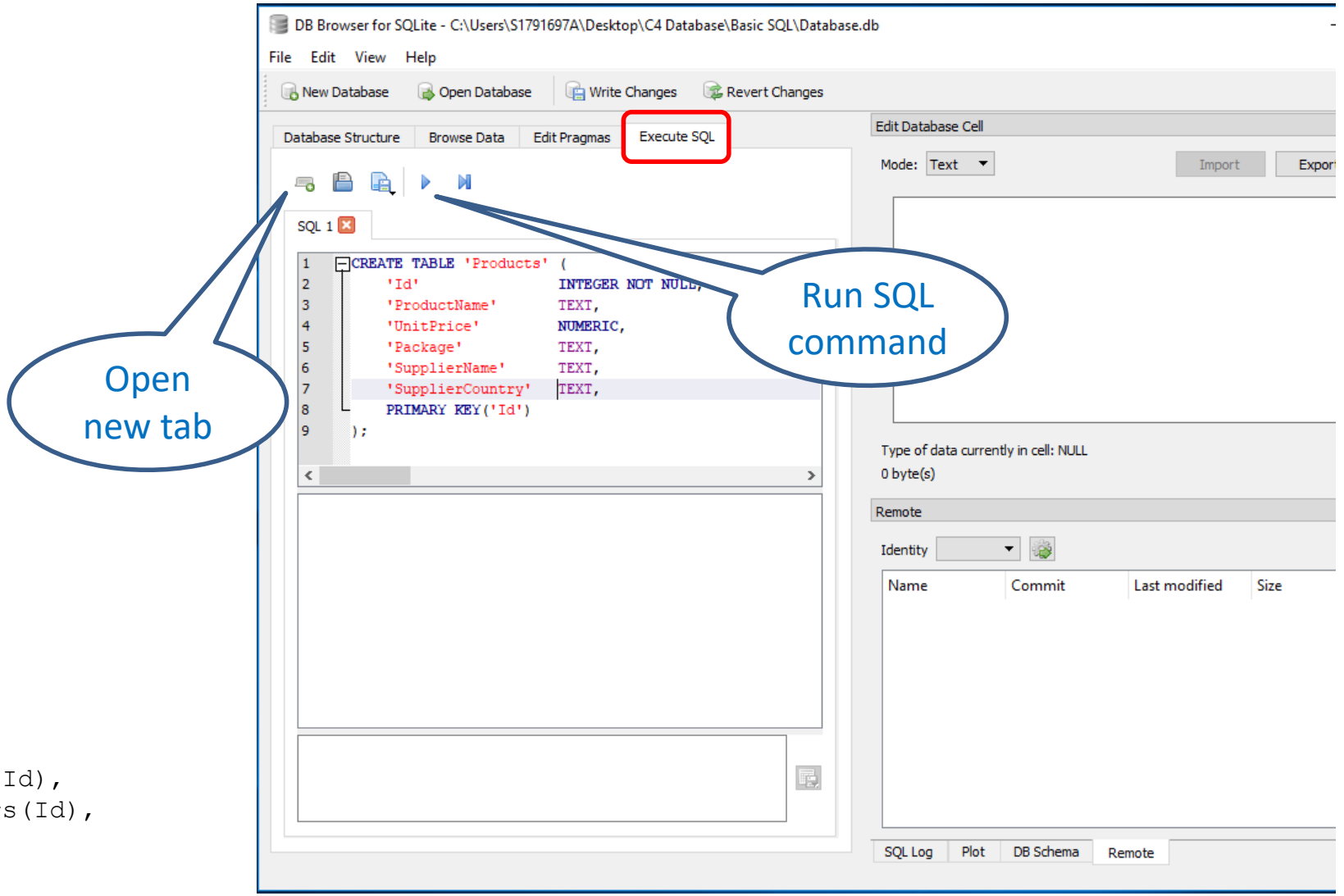
- Copy the create statements, paste them in a notepad and save it as "`Create Tables.sql`"



Execute SQL

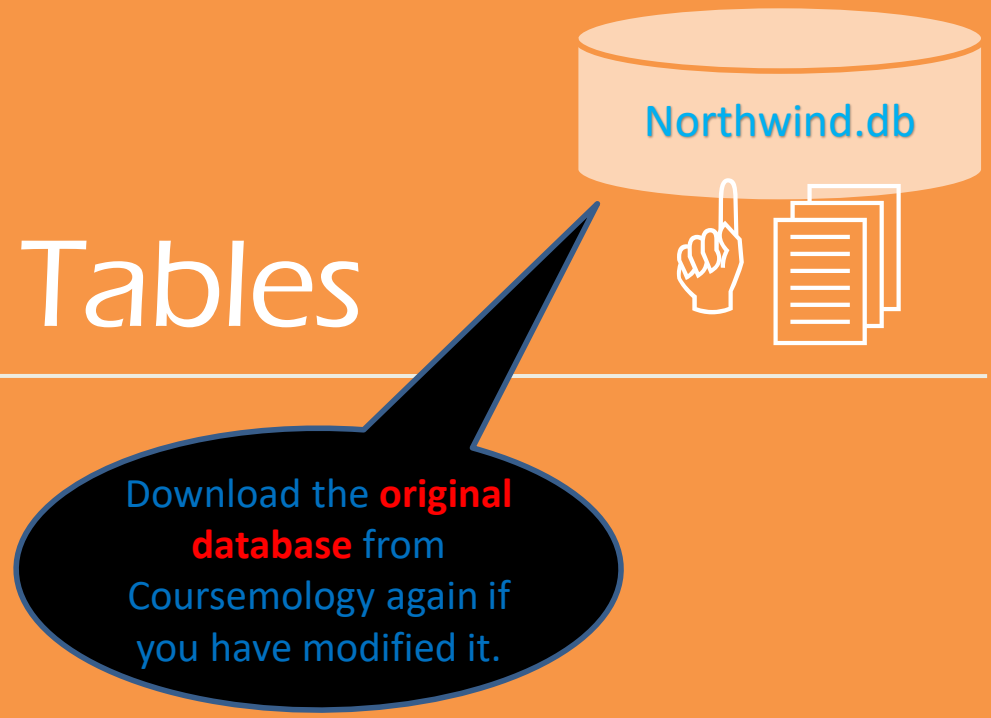
Create Tables using SQL Commands

```
CREATE TABLE Products (
Id                  INTEGER NOT NULL,
ProductName         TEXT,
UnitPrice           REAL,
Package             TEXT,
SupplierName        TEXT,
SupplierCountry     TEXT,
PRIMARY KEY(Id)
);
CREATE TABLE Customers (
Id                  INTEGER NOT NULL,
FirstName           TEXT NOT NULL,
LastName            TEXT NOT NULL,
City                TEXT,
Country             TEXT NOT NULL,
Phone               TEXT NOT NULL,
PRIMARY KEY(Id)
);
CREATE TABLE Orders (
Id                  INTEGER NOT NULL,
OrderDate           TEXT NOT NULL,
CustomerID          INTEGER NOT NULL,
ProductID           INTEGER NOT NULL,
Quantity            INTEGER NOT NULL,
FOREIGN KEY(ProductID) REFERENCES Products(Id),
FOREIGN KEY(CustomerID) REFERENCES Customers(Id),
PRIMARY KEY(Id)
);
```



C4-4b: Query Multiple Tables

Advanced SQL



Entity Relation (ER) Diagram



Linked Entity

Table Description :

Customers (Id, FirstName, LastName, City, Country, Phone)

Products(Id, ProductName, UnitPrice, Package, SupplierName, SupplierCountry)

Orders (Id, OrderDate, CustomerID*, ProductID*, Quantity)

__ : Primary Key;

* : Foreign Key

Challenge #1:

How to find the Customers, i.e. FirstName and LastName, who purchased the product "Lakkalik ri" in Feb 2013?

Problems:

- If we look into the Orders table and search for all the orders made in Feb 2013, we will find all the orders, but which ProductID corresponds to "Lakkalik ri"?
- Let's say we know the ProductID for "Lakkalik ri", but the Orders table will only provide us the CustomerIDs, how can we find their names?

Query involving ONE tables

Step 1:

- Find the ProductID for the product "Lakkalik ri".

```
SELECT Id FROM Products
WHERE ProductName = "Lakkalik ri";
```

Products table

	Id	ProductName	UnitPrice	Package	SupplierName	SupplierCountry
1	76	Lakkalik ri	18	500 ml	Karkki Oy	Finland

Query involving ONE table

Step 2:

- Find the CustomerIDs who purchased ProductID=76 in Feb 2013.

```
SELECT CustomerID, ProductID, OrderDate
FROM Orders
WHERE OrderDate LIKE "%Feb%2013%"
AND ProductID = 76;
```

	CustomerID	ProductID	OrderDate
1	24	76	Feb 3 2013 12:00:00:000AM
2	87	76	Feb 5 2013 12:00:00:000AM

Query involving ONE table

Step 3:

- Find the FirstName and LastName of the customers with Id=24 and Id=87.

```
SELECT FirstName, LastName FROM Customers WHERE
Id=24 or Id=87;
```

	Id	FirstName	LastName
1	24	Maria	Larsson
2	87	Pirkko	Koskitalo

Challenge #1 (con't):

How to find the Customers, i.e. FirstName and LastName, who purchased the product "Lakkalik ri" in Feb 2013?

One possible solution:

- If we know that the ProductID for "Lakkalik ri" is 76, then Step 2 and 3 can be combined together.

Query involving TWO tables

- Find the Customer, i.e. FirstName and LastName, who purchased ProductID=76 in Feb 2013.

Orders Table

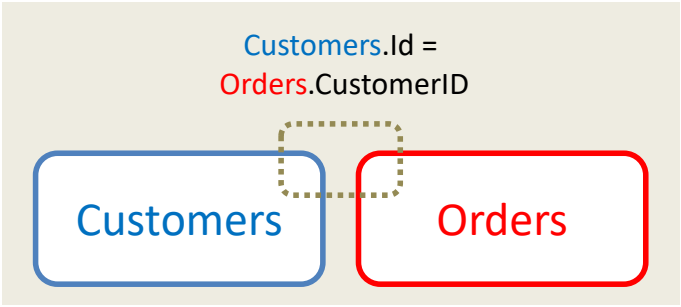
❶

	Id	OrderDate	CustomerID	ProductID	Quantity
1	187	Feb 3 2013 12:00:00:000AM	24	76	6
2	190	Feb 5 2013 12:00:00:000AM	87	76	30

Customers Table

❷

	Id	FirstName	LastName	City	Country	Phone
1	24	Maria	Larsson	Br cke	Sweden	0695-34 67 21
2	87	Pirkko	Koskitalo	Oulu	Finland	981-443655



Query involving TWO tables

- Find the Customer, i.e. FirstName and LastName, who purchased ProductID=76 in Feb 2013.

```
SELECT Customers.Id, FirstName, LastName
FROM Orders, Customers
WHERE OrderDate LIKE "%Feb%2013%"
AND ProductID=76
AND Customers.Id = Orders.CustomerID;
```

Since the field name CustomerID only appear in the Orders table, so it is unnecessary to specify the table name.

No need to write `Orders.CustomerID`

	Id	FirstName	LastName
1	24	Maria	Larsson
2	87	Pirkko	Koskitalo

Challenge #1 (con't):

How to find the Customers, i.e. FirstName and LastName, who purchased the product "Lakkalik ri" in Feb 2013?

The best solution:

- Can we combine Step 1, 2 and 3 together to perform a single query?

Query involving THREE tables

How to find the Customers, i.e. FirstName and LastName, who purchased the product "Lakkalik ri" in Feb 2013?

Orders Table

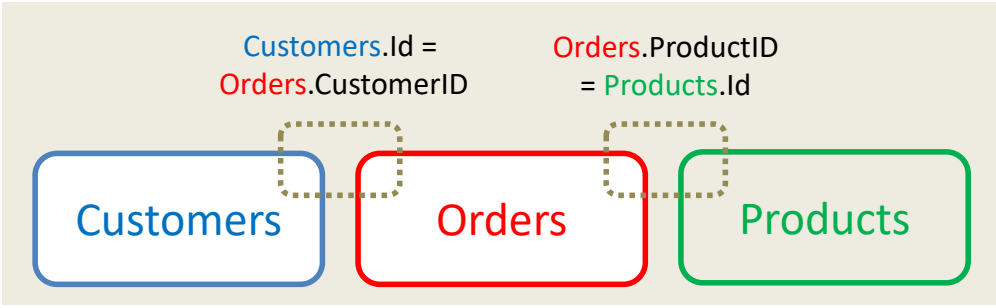
	Id	OrderDate	CustomerID	ProductID	Quantity
1	187	Feb 3 2013 12:00:00:000AM	24	76	6
2	190	Feb 5 2013 12:00:00:000AM	87	76	30

Products table

	Id	ProductName	UnitPrice	Package	S
1	76	Lakkalik ri	18	500 ml	Ka

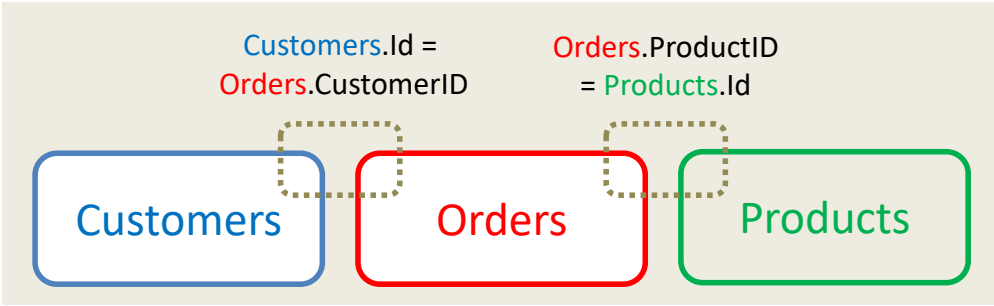
Customers Table

	Id	FirstName	LastName	City	Country	Phone
1	24	Maria	Larsson	Br cke	Sweden	0695-34 67 21
2	87	Pirkko	Koskitalo	Oulu	Finland	981-443655



Query involving THREE tables

How to find the Customers, i.e. FirstName and LastName, who purchased the product "Lakkalik ri" in Feb 2013?




```
SELECT Customers.Id, FirstName, LastName
FROM Orders, Customers, Products
WHERE OrderDate LIKE "%Feb%2013%"
AND ProductName = "Lakkalik ri"
AND Customers.Id = Orders.CustomerID
AND Orders.ProductID = Products.Id;
```

	Id	FirstName	LastName
1	24	Maria	Larsson
2	87	Pirkko	Koskitalo

No need to write `Orders.ProductID`

No need to write `Orders.CustomerID`

Challenge #2:

How to find how much Manuel Pereira spent on Jul 30 2012?

Problems:

- If we to look into the Orders table, how can we know which were Manuel Pereira's orders if we don't know his CustomerID?
- Let's say we now know his CustomerID, but how do we know which product cost how much? If not, how can we add the total?

Combining multiple queries

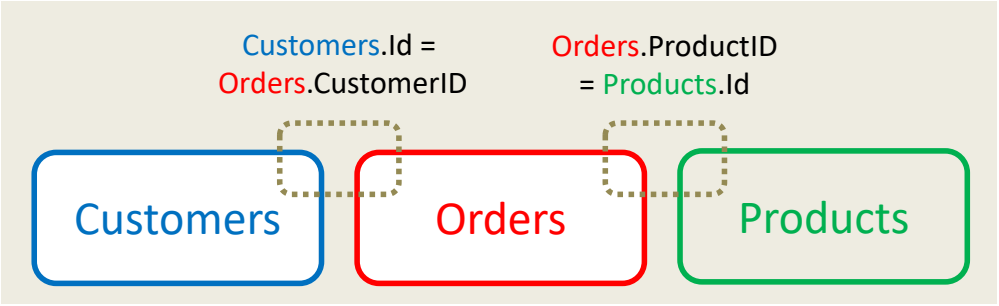
What were the products ordered by **Manuel Pereira**?

1
```
SELECT * FROM Customers
WHERE FirstName='Manuel' AND LastName='Pereira';
```

Id	FirstName	LastName	City	Country	Phone
33	Manuel	Pereira	Caracas	Venezuela	(2) 283-2951

2
```
SELECT * FROM Orders
WHERE CustomerId = 33;
```

Id	OrderDate	CustomerID	ProductID	Quantity
21	Jul 30 2012 12:00:00:000AM	33	2	20
538	Dec 18 2013 12:00:00:000AM	33	54	6



3
```
SELECT * FROM Products
WHERE Id=2 or Id=54;
```

Id	ProductName	UnitPrice	Package	SupplierName	SupplierCountry
2	Chang	19	24 - 12 oz bottles	Exotic Liquids	UK
54	TourtiSre	7.45	16 pies	Ma Maison	Canada

```
SELECT ProductName FROM Customers, Products, Orders
WHERE FirstName='Manuel' AND LastName='Pereira'
AND Customers.Id=Orders.CustomerID
AND Orders.ProductID=Products.Id;
```

	ProductName
1	Chang
2	TourtiSre

Combining multiple queries

How much did **Manuel Pereira** spend on **Jul 30 2012**?

❶
```
SELECT * FROM Customers
WHERE FirstName= 'Manuel' AND LastName='Pereira';
```

Id	FirstName	LastName	City	Country	Phone
33	Manuel	Pereira	Caracas	Venezuela	(2) 283-2951

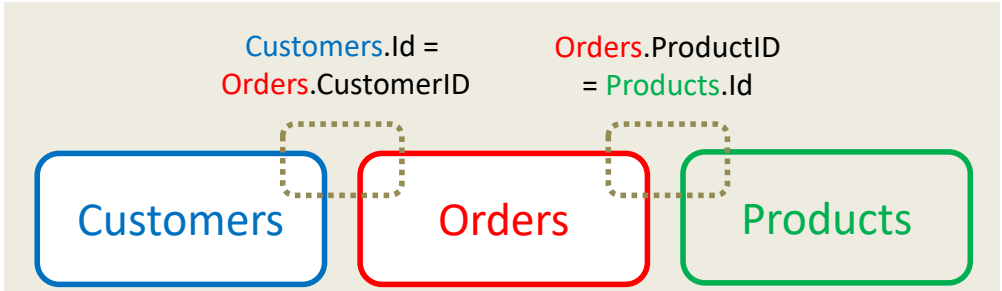
❷
```
SELECT * FROM Orders
WHERE CustomerId = 33
AND OrderDate LIKE 'Jul 30 2012%';
```

Id	OrderDate	CustomerID	ProductID	Quantity
21	Jul 30 2012 12:00:00:000AM	33	2	20

OrderDate LIKE 'Jul 30 2012%'

❸
```
SELECT * FROM Products
WHERE Id=2;
```

Id	ProductName	UnitPrice	Package	SupplierName	SupplierCountry
2	Chang	19	24 - 12 oz bottles	Exotic Liquids	UK




```
SELECT Quantity*UnitPrice FROM Customers, Products, Orders
WHERE FirstName='Manuel' AND LastName='Pereira'
AND OrderDate LIKE 'Jul 30 2012%'
AND Customers.Id=Orders.CustomerID
AND Orders.ProductID=Products.Id;
```

UnitPrice*Quantity
380

Note:

```
SELECT ProductName FROM Customers, Products, Orders
WHERE FirstName='Manuel' AND LastName='Pereira'
AND Customers.Id=Orders.CustomerID
AND Orders.ProductID=Products.Id;
```

No need to write `Customers.FirstName`

No need to write `Orders.CustomerID`

	ProductName
1	Chang
2	TourtiSre

```
SELECT UnitPrice*Quantity FROM Customers, Products, Orders
WHERE FirstName='Manuel' AND LastName='Pereira'
AND OrderDate LIKE 'Jul 30 2012%'
AND Customers.Id=Orders.CustomerID
AND Orders.ProductID=Products.Id;
```

No need to write `Products.UnitPrice * Orders.Quantity`

UnitPrice*Quantity
380

- Some field names appear only in their respective tables, so it is unnecessary to specify the table names. Eg. `Customers.FirstName`, `Orders.Quantity`, `Orders.ProductID`, `Orders.CustomerID`, `Products.ProductName` …, we can simply write without specifying the table names.

- Since '`Id`' is used in all the 3 tables, so it is necessary to specify the table names. Eg. `Customers.Id` and `Products.Id`

The End

C4 Database Concept and Basic SQL

Formula Sheet for Exam

- Create Table
- Query
- Insert data
- Update data
- Delete data
- Drop Table
- Types, Constraints, Functions and Operators

2 SQL Statements

```
CREATE TABLE table_name(
    column1_name COLUMN1_TYPE COLUMN1_CONSTRAINTS,
    column2_name COLUMN2_TYPE COLUMN2_CONSTRAINTS,
    ...
    PRIMARY KEY (column1_name, column2_name, ...),
    FOREIGN KEY (column_name) REFERENCES table_name(column_name)
);
```

```
SELECT column1_name, column2_name, ...
FROM table_name
WHERE where_expression
ORDER BY order_expression ASC;
```

```
SELECT column1_name, column2_name, ...
FROM table_name
WHERE where_expression
ORDER BY order_expression DESC;
```

```
SELECT table1_name.column1_name, table2_name.column2_name, ...
FROM table_name, table2_name
WHERE where_expression;
```

```
SELECT table1_name.column1_name, table2_name.column2_name, ...
FROM table1_name
INNER JOIN table2_name ON join_expression;
```

```
SELECT table1_name.column1_name, table2_name.column2_name, ...
FROM table1_name
LEFT OUTER JOIN table2_name ON join_expression;
```

```
SELECT
    COUNT(*),
    MAX(column1_name),
    MIN(column2_name),
    SUM(column3_name),
    ...
FROM table_name;
```

```
INSERT INTO table_name(column1_name, column2_name, ...)
VALUES(column1_value, column2_value, ...);
```

```
UPDATE table_name SET
    column1_name = column1_expression,
    column2_name = column2_expression,
    ...
WHERE where_expression;
```

```
DELETE FROM table_name
WHERE where_expression;
```

```
DROP TABLE table_name;
```

3 SQLite Types, Constraints, Functions and Operators

Types	Constraints	Functions	Operators			
NULL	NOT NULL	COUNT()	||	/	<	AND
REAL	PRIMARY KEY	MAX()	+	%	<=	OR
INTEGER	AUTOINCREMENT	MIN()	-	=	>	IS
TEXT	UNIQUE	SUM()	*	!=	>=	IS NOT